

A Comparative Analysis of Patient Similarity Measures for Outcome Prediction

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Abstract

Personalized medicine aims to improve clinical outcomes by tailoring treatments to individual patients based on genetic, phenotypic, or psychosocial characteristics, leveraging insights from similar patients. This is particularly necessary for managing diseases with significant variability in their causes, progressions and prognoses. Accurate measurement of patient similarity is crucial in this context, as it enables the identification of a high-quality cohort of similar patients, thereby enhancing clinical decision making with better evidence. However, previous studies have not comprehensively compared different patient similarity measures in large-scale retrospective analyses of electronic health records (EHRs). In this study, we conducted a comparative analysis of four patient similarity measures focusing on feature weighting mechanisms, using EHR data from 46,968 hospitalized patients. For evaluation, we assessed the patient similarity measures for predicting acute kidney injury, readmission, and mortality. Our results showed that using grid-searched weights to combine features based by their types outperformed all other methods.

Introduction

As electronic health record (EHR) systems evolve and become more integral to healthcare, they generate vast amounts of patient data, paving the way for personalized medicine. Personalized medicine involves making intelligent and individualized decisions on treatment plans and forecasting potential clinical outcomes by leveraging information from similar patients¹. Physicians can apply case-based reasoning on a cohort of similar patients, allowing them to predict disease onset, develop personalized treatment plans based on prior similar cases, and anticipate potential prognoses. Moreover, personalized prediction models built on similar patient cohorts are often more accurate, whereas traditional one-size-fits-all approaches often fail to account for variability in patient populations².

Patient similarity estimation plays a critical role in personalized medicine because it directly affects the quality of patient cohorts used for case-based reasoning and predictive modeling³. Among the various approaches to calculating patient similarity, distance-based measures are the most widely utilized. In these methods, each patient's EHR is represented as a vector, and similarity between vectors are calculated using different algorithms. Distance-based similarity measures can be further categorized into two types: basic-distance-based measures and Mahalanobis-distance-based measures.

Basic-distance-based measures, such as Euclidean distance, cosine distance, and the Jaccard index, do not require learning feature space transformations from data. For instance, David et al. demonstrated the effectiveness of employing the Euclidean distance on weighted features to select similar patients for an index patient⁴. Instead of weighting individual features, Wang et al. explored weighting features types (e.g. demographics, laboratory tests, diagnoses) to calculate patient similarity^{5, 6}. Fang et al. proposed an expert-based weighting system, where clinical experts assigned weights to features based on their perceived importance, leading to consensus-derived weights for use in similarity estimation⁷. Hielscher et al. suggested subgrouping the patient population by gender before applying a k-nearest neighbor (k-NN) method, showing that tailored predictors can improve accuracy⁸. Park et al. investigated the optimal cohort size in a Euclidean k-NN setting⁹, while Liu et al. used the Euclidean distance to retrieve similar patient cohorts and employed transfer learning to enhance personalized predictive models².

In contrast, Mahalanobis-distance-based measures involve learning a linear transformation of the feature space from data. Ng et al. introduced Locally Supervised Metric Learning (LSML), which learns a Mahalanobis distance specific to a disease¹⁰. Wang et al. studied the effect of integrating multiple independently learned Mahalanobis metrics with varying degrees of importance¹¹.

Despite the diversity of patient similarity methods, their impact on cohort quality remains unclear. Additionally, many of these methods have been evaluated only on small datasets, raising questions about their performance on larger real-world datasets. In this study, we compared four basic-distance-based patient similarity measures, with a focus on different feature weighting strategies, for predicting patient outcomes using a large real-world retrospective EHR

dataset. We emphasized on feature weighting in basic-distance-based measures for two key reasons: (1) basic-distance-based measures are more interpretable and computationally efficient than Mahalanobis-distance-based approaches, making them more clinically applicable, and (2) feature weighting is one of the most commonly used techniques in designing novel basic-distance-based similarity measures.

To comprehensively evaluate the impact of various feature weighting strategies on cohort quality, we assessed them on three binary prediction targets: Acute Kidney Injury (AKI) onset, 30-day readmission, and 1-year mortality. These outcomes represent different clinical aspects: disease onset, short-term outcomes, and long-term outcomes. AKI is a life-threatening and highly heterogeneous condition prevalent in hospitalized patient population, where a personalized approach is particularly important. This study establishes a foundational understanding of how different feature weighting strategies in basic-distance-based patient similarity measures affect the cohort quality, offering valuable insights for future developments in patient similarity estimation.

Methods

1 Data Source and Processing

We extracted de-identified EHR data from patients hospitalized at the University of Kansas Medical Center (KUMC) for two or more days between January 1, 2015, and December 31, 2019. Patients were excluded if they met either of the following criteria: (1) AKI onset occurred within the first 2 days after admission, as the focus was on hospital-acquired AKI¹², or (2) baseline serum creatinine (SCr) >3.5 mg/dL, indicating moderate-to-severe kidney dysfunction upon admission. We collected four types of clinical features, namely demographics, comorbidities, laboratory tests, and procedures, as well as records of 30-day readmission and 1-year mortality. Three target outcomes – AKI onset, 30-day readmission, and 1-year mortality – were used to evaluate the quality of similar patient cohorts constructed based on different similarity measures.

For AKI onset prediction, the risk prediction point was set as 1-day before onset for patients with AKI and 1-day before the last SCr measurement for patients without AKI. For laboratory tests, we used only the most recent measurement within this time frame, expressed as continuous values normalized between 0 and 1. Comorbidities and procedures were expressed as binary values. Demographics were binarized as follows: age over 60, male gender, and black race were assigned a value of 1, while all others were assigned a value of 0. This process resulted in over 10,000 unique variables in the feature space. To reduce the dimensionality, we excluded laboratory tests with >30% missing data and imputed missing values using Multiple Imputation by Chained Equations (MICE)¹³. For comorbidities and procedures, we applied chi-square tests to select the top 50 features from each feature type.

2 Patient Similarity Measures

To evaluate the quality of patient cohorts retrieved by different similarity measures, we established three prediction targets: AKI onset, 30-day readmission, and 1-year mortality. Each target was formulated as a standard k-NN binary classification task, with the label for each index patient determined by majority voting of their nearest neighbors. We compared 4 basic-distance-based patient similarity measures with a focus on different feature weighting strategies: (1) vanilla Euclidean distance; (2) weighting features by kidney-function-associated factors with empirically defined weights; (3) weighting features by types with empirically defined weights and (4) weighting features by types with grid-searched weights. We tested these measures under various parameter settings. Since the code for most of these studies is not publicly available, we implemented them based on descriptions in the literature and made necessary adaptations to fit our dataset and ensure fair comparisons within a unified framework. Random sampling was used as a baseline method. A summary of these methods is provided in **Table 1**.

Table 1. Summary of the benchmarked patient similarity measures.

Similarity measure	Source literature	Similarity Calculation	Feature Weighting
Random sampling	Liu et al. ²	The Euclidean distance	Equal weights
Vanilla Euclidean distance	Liu et al. ²		Equal weights
Weighting features by kidney-function-associated factors with empirically defined weights	Wang et al. ⁵	The Euclidean distance was used for laboratory tests and the Jaccard	Renal features taking a weight of 0.2, 0.5 and 0.8, respectively

Weighting feature by types with empirically defined weights	Fang et al. ⁷	index was used for demographics, comorbidities and procedures.	Demographics: 0.2, comorbidities: 0.2, laboratory tests: 0.3, procedures: 0.3
Weighting features by types with grid-searched weights	Wang et al. ⁶		Grid searching for the best weights to combine 2, 3 and 4 top-performing feature types, respectively

2.1 Random Sampling and Vanilla Euclidean Distance

Random sampling was chosen as the baseline for our experiments. A patient distance matrix was randomly generated with values between 0 and 1. Following min-max normalization, the patient similarity matrix was derived by subtracting each value in the normalized distance matrix from 1, as described in **Equation 1** and **2**. The resulting patient similarity matrix was then sorted in descending order to create the patient neighborhood matrix, which represents the similarity between each patient and others.

$$D'(i, j) = \frac{D(i, j) - \min(D)}{\max(D) - \min(D)} \quad \text{Equation 1}$$

$$S(i, j) = 1 - D'(i, j) \quad \text{Equation 2}$$

D represents the distance matrix, D' represents the normalized distance matrix, $D(i, j)$ represents the distance between the i -th patient and the j -th patient, and S represents the patient similarity matrix.

Vanilla Euclidean distance was used to calculate patient distances with uniform weights (**Equation 3**). From the resulting patient distance matrix, we then derived the patient similarity matrix and the neighbor matrices.

$$D_{Euc}(i, j) = \sqrt{(f_1^i - f_1^j)^2 + (f_2^i - f_2^j)^2 + \dots + (f_m^i - f_m^j)^2} \quad \text{Equation 3}$$

f_m^i represents the m -th feature of the i -th patient.

2.2 Weighting features by type with empirically defined weights

EHR data encompass various feature types, such as demographics and laboratory measurements. To account for their varying importance, Wang et al. constructed separate patient similarity matrices for each feature type and combined them through a weighted sum using empirically defined weights⁵. Their study assigned weights of 0.1, 0.1, 0.4, and 0.4 to age, sex, lab tests, and diagnoses, respectively. Following a similar approach, we assigned weights of 0.2, 0.2, 0.3, 0.3 to demographics, comorbidities, procedures and laboratory tests, respectively. We used the Jaccard index (**Equation 4**) to measure the similarity of demographics, comorbidities and procedures since they were binary, while using Euclidean distance for laboratory tests. The weighted sum is shown in **Equation 5**.

$$S_{Jaccard}(i, j) = \frac{|F^i \cap F^j|}{|F^i \cup F^j|} \quad \text{Equation 4}$$

$$S_{ft}(i, j) = w_d \cdot S_d(i, j) + w_c \cdot S_c(i, j) + w_p \cdot S_p(i, j) + w_l \cdot S_l(i, j) \quad \text{Equation 5}$$

F^i represents the binary feature space of the i -th patient, w_d, w_c, w_p, w_l and S_d, S_c, S_p, S_l represent the weights and patient similarity matrices of demographics, comorbidities, procedures and laboratory tests, respectively.

2.3 Weighting features by kidney-function-associated factors

Fang et al. proposed an expert-based feature weighting system for personalized models of type-2 diabetes mellitus, hypertension and dyslipidaemia⁷. Weights were determined based on the importance of each feature from consensus among a team of three clinicians and scaled accordingly. We adapted this method for AKI by assigning larger weights

to features associated with kidney function, which are presumed to have greater predictive value. The kidney-function-associated factors were selected from the remaining feature space based on literature¹⁴, including age, blood urea nitrogen, serum creatinine, estimated glomerular filtration rate, anion gap and chronic kidney disease (CKD) history. To make each patient similarity measure consistent and comparable within a unified framework, the patient similarity matrix for kidney-function-associated features was derived using Euclidean distance, while other features used the methods described in **Section 2.2**, scaled by their lengths to ensure equal weighting of each feature. The two patient similarity matrices were then combined using a weighted sum with the predefined weights, as illustrated in **Equation 6** and **7**. To explore effects of weights on model performance, we conducted experiments with kidney-function-associated features assigned weights of 0.2, 0.5, and 0.8, respectively.

$$w_k + w_{nk} = 1 \quad \text{Equation 6}$$

$$S_{kf}(i, j) = w_k \cdot S_k(i, j) + w_{nk} \cdot S_{nk}(i, j) \quad \text{Equation 7}$$

w_k , w_{nk} and S_k , S_{nk} represent the weights and patient similarity matrices of the kidney-function-associated features and the non-kidney-function-associated features, respectively.

2.4 Weighting features by type with grid-searched weights

Wang et al. proposed a grid-search approach to optimize feature type weights by evaluating predictive performance using individual feature types⁶. In this study, we applied grid-search to identify the best-performing feature types for predicting AKI onset, 30-day readmission, and 1-year mortality. The feature types considered were still demographics, comorbidities, laboratory tests and procedures. The top-performing feature types were then combined with optimized weights derived through grid-search. Their similarities were derived using the same methods described in **Section 2.2**. We conducted experiments using 2, 3, and all 4 feature types. For each prediction task, we optimized weights using AUPRC (area under the precision-recall curve) to emphasize performance on the positive class. Each task yielded a unique set of top-performing features and optimal weights. The detailed procedure is outlined in **Algorithm 1**.

Algorithm 1. Pseudocodes showing the steps of weighting features by types with grid-searched weights.

Patient similarity calculation

Input: training set, testing set

for $i = 1$ to number of feature types by 1:

$S_{\text{train}}^i, S_{\text{test}}^i \leftarrow$ Calculate feature type similarities for i -th feature type on training and testing set

end

for $i = 1$ to number of feature types by 1:

AUPRC $\leftarrow$ Evaluate k-NN model using only i -th feature type

End

$S_{\text{train}}^1, S_{\text{train}}^2 \leftarrow$ Select top-performing 2 (3, 4) feature type similarity matrices

$$(1)w_1 + w_2 = 1; (2)w_1 \geq w_2 > 0; (3)w_1 = 0.1m, w_2 = 0.1n, \quad m, n \in Z \quad \text{Equation 8-10}$$

while w_1, w_2 satisfy the constrain in **Equation 8-10** **do**:

$PS_{\text{train}} \leftarrow w_1 S_{\text{train}}^1 + w_2 S_{\text{train}}^2$

AUPRC $\leftarrow$ Evaluate k-NN model with PS_{train}

end while

$w_1^*, w_2^* \leftarrow$ Select w_1, w_2 with best performance

$PS_{\text{test}} \leftarrow w_1^* S_{\text{test}}^1 + w_2^* S_{\text{test}}^2$

return PS_{test}

3 Model evaluation and comparison

To ensure robust and reliable evaluation, we employed 100-fold cross-validation for testing each model. In each iteration, one fold was retained as the test set, while the remaining 99 folds served as the patient pool for retrieving similar patients. All models were evaluated on similar patient cohort sizes of 2%, 3%, 4% and 10% of the patient pool, respectively. The evaluation metrics used were area under the receiver operating characteristic curve (AUROC) and AUPRC for overall prediction performance. Additionally, we assessed the binary cross entropy (BCE) loss for evaluating prediction performance specific to the positive class. BCE loss, as shown in **Equation 11**, provides more granular insights into the internal consistency within the retrieved similar patient cohort.

$$Loss(y, p) = -y \log(p) - (1 - y) \log(1 - p) \quad \text{Equation 11}$$

y represents the true labels, and p represents the predicted probabilities.

To determine the significance of differences between models across the 100 simulations, we conducted one-tailed paired t-tests on the three metrics, with a significance threshold set at $p < 0.01$. The workflow of the study is shown in **Figure 1**.

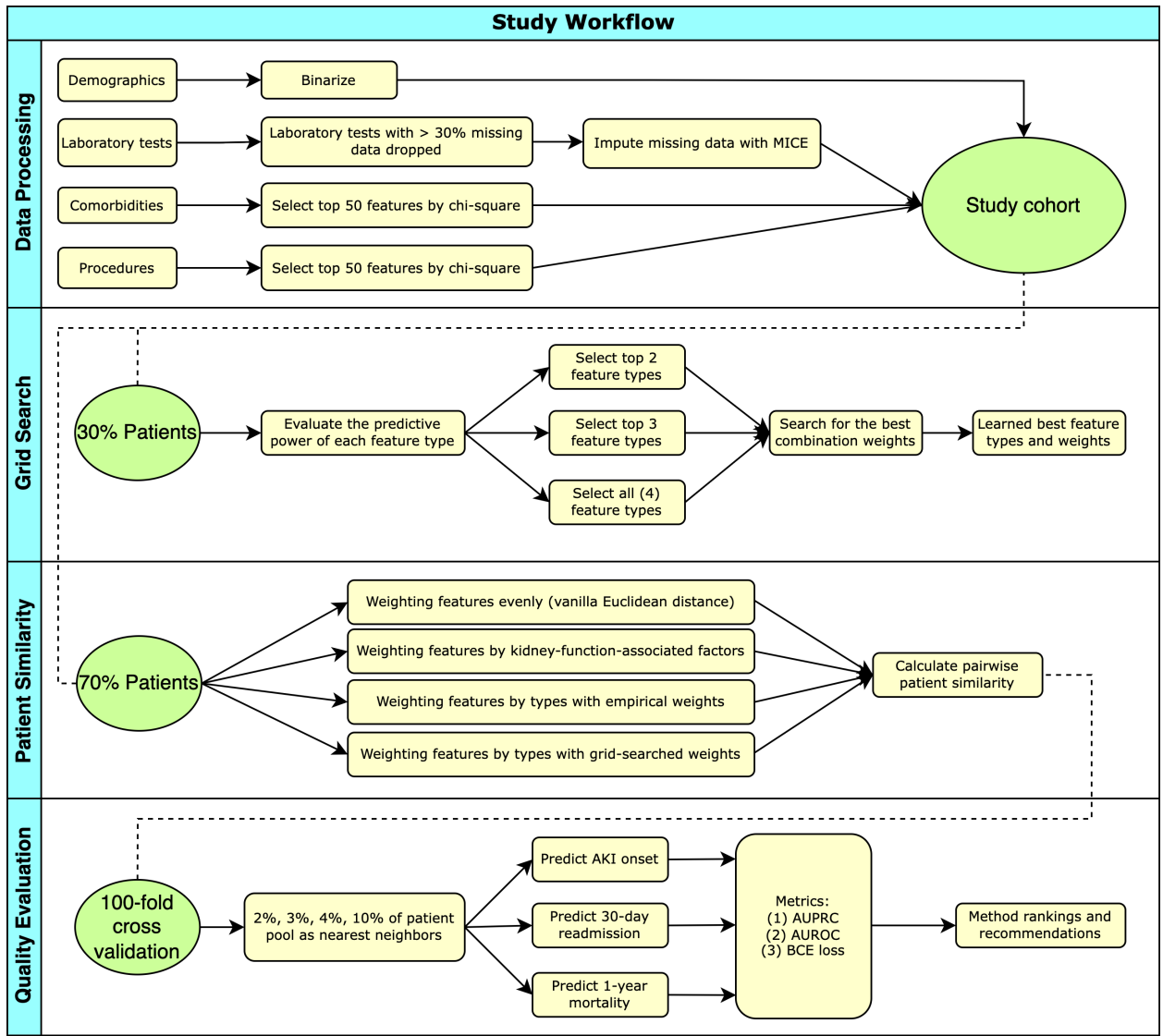


Figure 1: Workflow of the study.

Results and Discussion

The study cohort included 46,968 unique patients, with a median age of 62 years and a median hospital stay of 8 days. The overall incidence of AKI in the population was 10.92%, with more than 96% of initial AKI onset classified as stage-1. Detailed cohort characteristics are presented in **Table 2**.

Table 2. Characterization of the study cohort. The numbers in parentheses represent the 25th and 75th percentiles.

Number of unique patients	46,968
1. Demographics	
Age	62.0 (49.0-72.0)
Male	47.94%
White	76.8%
Black	13.96%
2. AKI information	
Length of hospital stay (days)	8 (4-20)
Overall AKI Rate	10.92%
AKI Stage 1	96.82%
AKI Stage 2	2.73%
AKI Stage 3	0.45%
3. AKI risk factors	
Hypertension	37.33%
Diabetes mellitus	16.07%
Heart failure	9.05%
Chronic kidney diseases	7.89%

The quality of similar patient cohorts retrieved by the benchmarked patient similarity measures, evaluated for three prediction targets – AKI onset, 30-day readmission, and 1-year mortality – is shown in **Figures 3, 4, and 5**, respectively. Each matrix represents evaluation results using AUPRC, AUROC, or BCE loss for specific cohort sizes. Each entry in the matrix indicates how often the method in the row outperformed the method in the column across 100 cross-validation iterations. The average rankings of each similarity measure are summarized in **Table 3**. The average rank displayed was derived from the original rankings of AUPRC, AUROC, and BCE loss over the cohort sizes of 3% (small cohort size) and 10% (large cohort size), reflecting the overall performance of each similarity measure.

For AKI onset prediction (**Figure 3**), grid search-based weighting significantly outperformed other methods small cohort sizes (2%-4%). This is intuitive, as pre-searching for optimal feature types and weights on the holdout dataset helps identify an optimal or near-optimal settings, resulting in strong performance on the test dataset. However, for larger cohort sizes (10%), only grid search weighting with all feature types significantly outperformed the vanilla Euclidean distance, while using only two or three feature types did not. Similar trends were observed for the other two prediction targets at the 10% cohort size (**Figure 4 and 5**). This suggest that incorporating more feature types may help prevent overfitting. Combining multiple feature types is essentially an ensemble approach, where each feature type contributes unique information to patient similarity, leading to more robust and accurate decision¹⁵.

Incorporating more feature types is particularly important when the holdout data is small because the ranking of top-performing features can overfit. For example, demographics might appear as most predictive in a small dataset, while laboratory tests could dominate in a larger dataset. This discrepancy explains why models relying on two or three top-performing feature types learned from small holdout datasets perform poorly on larger test datasets. Moreover, as the cohort size increases, there is higher likelihood of including patients from the opposite class (given the overall AKI rate of 10.92%) and incorporating multiple feature types helps the model maintain its performance.

When assigning larger weights to kidney function-related features, we observed a slight improvement over the Euclidean distance in AUPRC, and a significant improvement in AUROC. This indicates that incorporating expert knowledge can enhance the quality of the retrieved patient similarity cohorts. Notably, larger weight (0.8) outperformed smaller weights (0.2 and 0.5), suggesting that this expert knowledge informed strategy was effective for AKI onset prediction. However, it did not outperform grid search-based weighting.

When using BCE loss to evaluate the internal consistency of similar patient cohorts for positive patients in the test dataset, both the kidney-function-focused weighting and grid-search-based weighting significantly outperformed random sampling and vanilla Euclidean distance for small cohort sizes (2% and 3%). However, for larger cohort sizes (10%), only grid-search-based weighting maintained its advantage. This suggests that grid-search-based weighting is more effective in constructing high-quality similar patient cohorts for rare positive cases in highly imbalanced datasets.

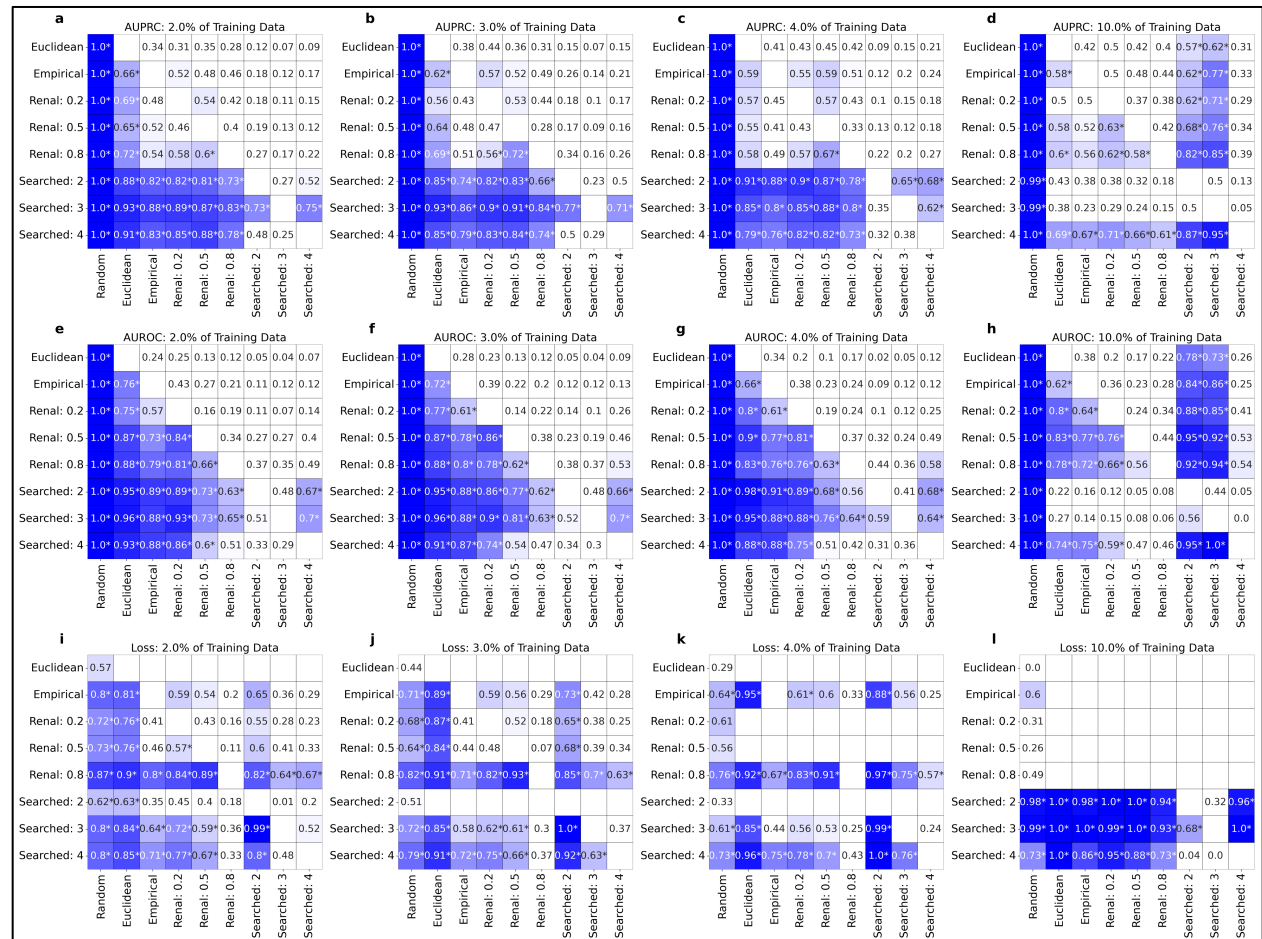


Figure 3. Heatmaps comparing patient similarity measure performance with AKI onset as the prediction target. Number in each cell is the percentage of times that the model in row had a higher performance than the model in column in the 100-fold cross validation. The asterisk represents statistically significant results by one-tailed paired t-test. The performance that was worse than random sampling is hidden.

For 30-day readmission prediction (**Figure 4**), using AUPRC and AUROC as evaluation metrics, empirical feature type weighting, kidney-function-focused weighting, and grid-search-based weighting performed comparably to vanilla Euclidean distance. The reason may be that readmission was not well predicted by the current feature space as reasons for readmission can vary greatly¹⁶. Even with the grid search method, it was challenging to find a set of weights that outperformed the vanilla Euclidean distance given the limited information available.

This highlights the importance of tailoring the feature space to the specific prediction target in personalized medicine to achieve optimal model performance. A good example of this is the performance of kidney-focused weighting with a weight of 0.8. In **Figure 4**, this method was the worst performer, likely because weights were all assigned to kidney

function-related features, which have poor predictive power for readmission due to various causes. Meanwhile, other more general and predictive features, such as age and cardiovascular diseases, had their weights significantly reduced, leading to the method's poor performance. When using BCE loss as the metric, grid-search-based weighting outperformed other methods across all cohort sizes, although their absolute performance was unsatisfactory compared to AUPRC and AUROC.



Figure 4. Heatmaps comparing patient similarity measure performance with 30-day readmission as the prediction target.

At last, for 1-year mortality prediction (**Figure 5**), kidney-function-focused weighting with a weight of 0.2, empirical-feature-type weighting, and grid-search-based weighting with all feature types performed best. The difference between kidney-function-focused weighting with a weight of 0.2 and empirical-feature-type weighting is minimal, with former assigns slightly higher weights to kidney-related features. Their similar performance can also be observed in **Figures 3 and 4**. The grid-search-based weighting did not significantly outperform the aforementioned two methods, likely due to weights learned from the holdout dataset not generalizing well to the test dataset. This issue could be mitigated by increasing the size of the holdout dataset. However, the grid-search-based weighting remained the best method in terms of BCE loss.

	Random	Euclidean	Empirical	Renal: 0.2	Renal: 0.5	Renal: 0.8	Searched: 2	Searched: 3	Searched: 4
AKI onset	8.33 (1.21)	7.67 (1.37)	4.83 (1.33)	5.67 (1.03)	5.17 (2.14)	2.50 (1.38)	4.83 (2.79)	3.50 (3.21)	2.50 (1.05)
30-day readmission	7.50 (2.35)	5.50 (3.02)	3.33 (1.03)	3.33 (2.34)	6.00 (1.26)	7.67 (0.82)	5.17 (2.93)	3.17 (2.56)	3.33 (1.37)

1-year mortality	8.50 (0.84)	5.50 (2.74)	2.00 (1.10)	3.17 (1.17)	5.33 (0.82)	6.83 (0.75)	6.00 (3.16)	5.33 (1.86)	2.33 (1.63)
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Table 3. Averaged rankings of each patient similarity measure across the three prediction tasks. The numbers in parentheses represent the standard deviations.

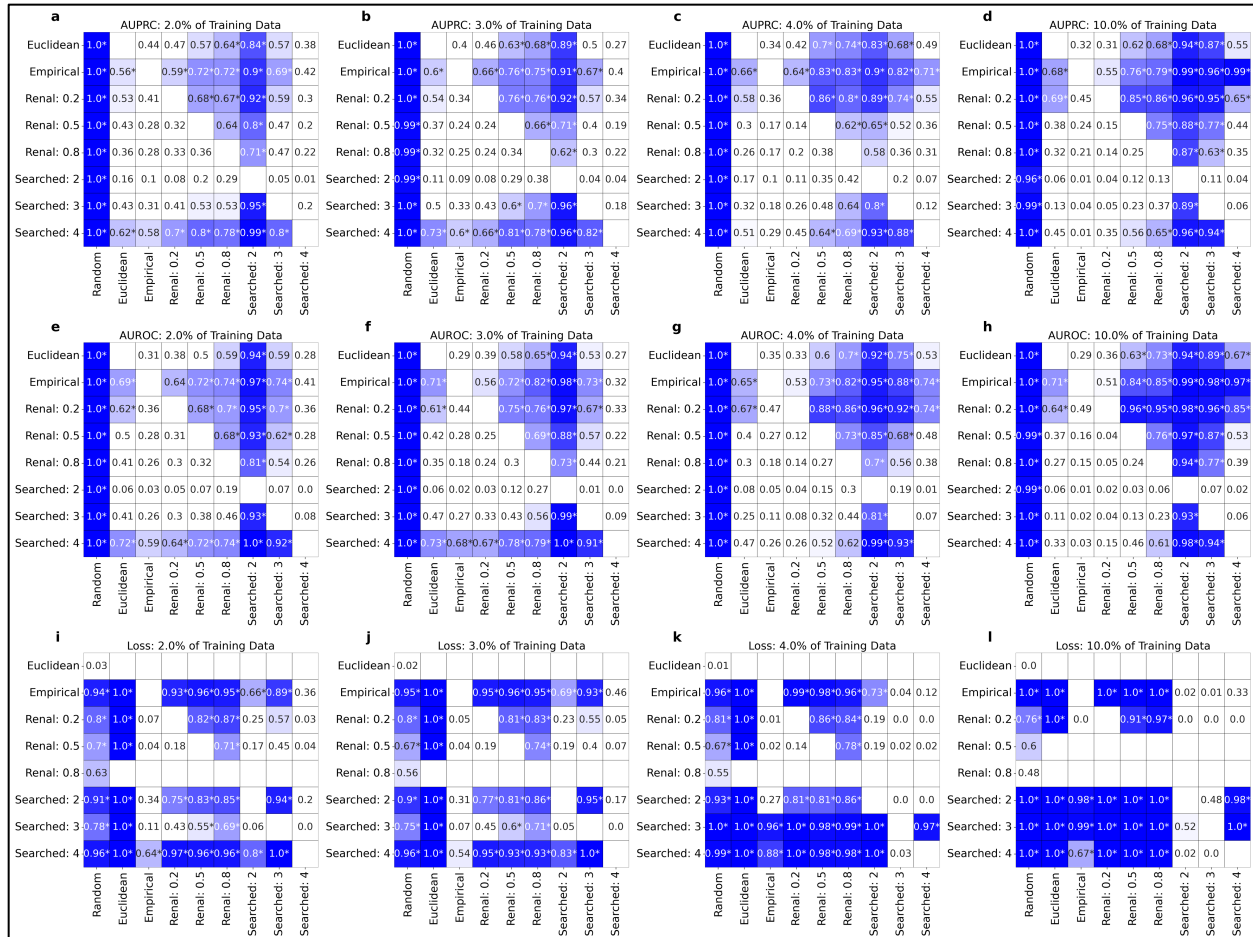


Figure 5. Heatmaps comparing patient similarity measure performance with 1-year mortality as the prediction target.

In summary, grid-search-based weighting with all feature types provided the highest or second-highest quality of similar patient cohorts across the three prediction targets, although it requires an additional holdout dataset for optimal weight search. If the holdout dataset is small or its distribution differs from the test population, there is a risk of poor generalization. Furthermore, this approach incurs higher computational costs. When resources are limited, expert-knowledge-informed weighting can be viable alternative.

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